

Solución

1.a

$$\begin{aligned} L(p) &= \prod_{i=1}^n p(1-p)^{x_i-1} \\ &= p^n (1-p)^{\sum_{i=1}^n x_i - n} \end{aligned}$$

10 pts

$$\rightarrow L(p) = n \ln(p) + \left(\sum_{i=1}^n x_i - n \right) \cdot \ln(1-p)$$

$$\rightarrow \frac{\partial L(p)}{\partial p} = n \cdot \frac{1}{p} + \left(\sum_{i=1}^n x_i - n \right) \frac{1}{1-p} \cdot (-1) = 0$$

$$\rightarrow \frac{n}{p} = \frac{\sum_{i=1}^n x_i - n}{1-p}$$

$$\rightarrow \frac{1-p}{p} = \frac{\sum_{i=1}^n x_i - n}{n}$$

10 pts

$$\rightarrow \frac{1}{p} = \bar{x}$$

$$\rightarrow \hat{p}_{ML} = \frac{1}{\bar{x}}$$

1.6

$$\hat{\theta}^* = E(\hat{\theta}_1 | U=u) \text{ por Th de Rao Blackwell}$$

4 pts

$$= 0 \cdot \cancel{p(\hat{\theta}_1=0 | U=u)} + 1 \cdot p(\hat{\theta}_1=1 | U=u)$$

$$= \frac{p(X_1=1, \sum_{i=1}^n X_i = u)}{p(\sum_{i=1}^n X_i = u)}$$

$$= \frac{p(X_1=1, \sum_{i=2}^n X_i = u-1)}{p(\sum_{i=1}^n X_i = u)}$$

10 pts

donde $\sum_{i=1}^n X_i \sim \text{Bin. Negativa}(n, p)$

y $\sum_{i=2}^n X_i \sim \text{Bin. Negativa}(n-1, p)$

$$\rightarrow \frac{\cancel{p} \binom{u-1-1}{n-1-1} \cdot \cancel{p}^{n-1} (1-\cancel{p})^{u-1-(n-1)}}{\binom{u-1}{n-1} \cdot \cancel{p}^n (1-\cancel{p})^{u-n}}$$

10 pts

$$\rightarrow \hat{\theta}^* = \frac{\binom{u-2}{n-2}}{\binom{u-1}{n-1}} ; \text{ donde } u = \sum_{i=1}^n X_i$$

$$\rightarrow \hat{\theta}^* = \frac{\frac{(v-2)!}{(\cancel{v-n})! \cdot (n-2)!}}{\frac{(v-1)!}{(\cancel{v-n})! \cdot (n-1)!}} = \frac{n-1}{v-1} = \frac{n-1}{\sum_{i=1}^n x_i - 1} \quad \text{1 pt}$$

2.a) $U = \ln(1 + e^{X-\mu})$

$$\rightarrow e^U = 1 + e^{X-\mu}$$

$$\text{Si } X \rightarrow -\infty \Rightarrow U = \ln(1) = 0$$

$$\rightarrow e^U - 1 = e^{X-\mu}$$

$$\text{Si } X \rightarrow +\infty \Rightarrow U = \ln(+\infty) = +\infty$$

$$\rightarrow \ln(e^U - 1) = X - \mu$$

$$\rightarrow \underbrace{\ln(e^U - 1) + \mu}_{g^{-1}} = X$$

g^{-1}

$$\bullet \frac{\partial g^{-1}}{\partial U} = \frac{e^U}{e^U - 1}$$

$$f_U = \frac{e^{-[\ln(e^U - 1) + \cancel{\mu} - \cancel{\mu}]}}{\left(1 + e^{-[\ln(e^U - 1) + \cancel{\mu} - \cancel{\mu}]}\right)^2} \cdot \frac{e^U}{e^U - 1}$$

$$= \frac{e^{-\ln(e^U - 1)}}{\left(1 + e^{-\ln(e^U - 1)}\right)^2} \cdot \frac{e^U}{e^U - 1} \quad ; \quad \text{Si } U > 0$$

$$= \frac{e^{\ln\left(\frac{1}{e^U - 1}\right)}}{\left[1 + e^{\ln\left(\frac{1}{e^U - 1}\right)}\right]^2} \cdot \frac{e^U}{e^U - 1} \quad ; \quad \text{Si } U > 0$$

10 pts

$$= \frac{\frac{1}{e^v - 1} \cdot \frac{e^v}{e^v - 1}}{\left(1 + \frac{1}{e^v - 1}\right)^2} \quad \text{Si } v > 0$$

$$= \frac{\frac{e^v}{\cancel{(e^v - 1)^2}}}{\left[\frac{\cancel{e^v} - \cancel{1} + 1}{\cancel{e^v - 1}}\right]^2} \quad \text{Si } v > 0$$

$$= \frac{e^v}{e^{2v}} = e^{-v} \quad \text{Si } v > 0$$

$$\therefore f_v = \int_0^\infty e^{-v} \quad \text{Si } v > 0$$

e.o.c

$$\therefore v \sim \text{Exp}(1)$$

10 pts

2.b

Note que $U = \ln(1 + e^{x-\mu})$ es una
función de los datos y del parámetro μ

5 pts

Además $f_U \sim \text{Exp}(1)$ la cual no depende
de μ .

5 pts

Por ende U es un pivote para μ

$$(2.c) \quad P(a < U < b) = 0.95$$

$$\rightarrow \int_0^a e^{-u} = 0.025$$

$$\rightarrow -e^{-u} \Big|_{u=0}^{u=a} = 0.025$$

$$\rightarrow -e^{-a} + 1 = 0.025$$

$$\rightarrow e^{-a} = 0.975$$

$$\rightarrow a = -\ln(0.975)$$

De manera análoga $b = -\ln(0.025)$

5 pts

$$\rightarrow P(a < \ln(1 + e^{x-\mu}) < b)$$

$$\rightarrow P(e^a < 1 + e^{x-\mu} < e^b)$$

$$\rightarrow P(e^a - 1 < e^{x-\mu} < e^b - 1)$$

$$\rightarrow P(\ln(e^a - 1) < x - \mu < \ln(e^b - 1))$$

$$\rightarrow P(\ln(e^a - 1) - x < -\mu < \ln(e^b - 1) - x)$$

$$\rightarrow P(-\ln(e^a - 1) + x > \mu > -\ln(e^b - 1) + x)$$

10 pts

donde $a = -\ln(0.975)$ y $b = -\ln(0.025)$

Por ende el I_C bilateral del 95% es

$$\left[-\ln\left(e^{-\ln(0.025)} - 1\right) + x \ ; \ -\ln\left(e^{-\ln(0.975)} - 1\right) + x \right]$$

5 pts

$$\approx \left[-3.66 + x \ ; \ 3.66 + x \right]$$

2.d) Si se estima el IC del 95% con el valor de $\bar{X}$ se obtiene

$$\sim [5,69 ; 13,02]$$

Por ende un valor de $\mu = 6$ es razonable

con un 95% de confianza, pues cae dentro del intervalo estimado.

5 pts

3.a) Aplicando el Método Delta.

$$\text{Note que } g(x) = \frac{1}{\sqrt{x}} \rightarrow g'(x) = -\frac{1}{2} \cdot \frac{1}{x^{3/2}}$$

→

$$\sqrt{n} \cdot \left(\frac{1}{\sqrt{\bar{x}}} - \frac{1}{\sqrt{\theta}} \right) \xrightarrow{d} N\left(0, \theta^4 \cdot \left(-\frac{1}{2} \cdot \frac{1}{\theta^{3/2}}\right)^2\right)$$

$$\xrightarrow{d} N\left(0, \frac{\theta}{4}\right)$$

6 pts

→

$$\frac{1}{\sqrt{\bar{x}}} \xrightarrow{d} N\left(\frac{1}{\sqrt{\theta}}, \frac{\theta}{4n}\right)$$

4 pts.

//